

1: Laptop Hardware & Mobile Devices

Part 1: Laptop Hardware - Components, Displays & Connectivity

!? Question 1 - What are the differences between a HDD (magnetic disk) and a SSD (solid state disk)? Why might you choose one over the other?

The difference is that they use different media to store data. HDD stores data on spinning magnetic platters by making small bumps on the platters, and uses the head to read them, SDD has a collection of chips with no moving parts. SDD will often perform better than HDD.

!? Question 2 - Why might a system administrator choose to image and clone a disk when replacing a hard drive?

So you don't have to manually move files to the new SSD.

!? Question 3 - Identify the labeled components in the figure below:

- Component A: Hard disk drive
- Component B:
- Component C: M.2 SSD
- Component D: RAM
- Component E:



Component A



Component B



Component C



Component D



Component E

! Question 4 - What advantages does using biometric authentication have over using a password?

Biometrics are unique

! Question 5 - Describe the advantages and disadvantages of the following types of display technologies:

- In-plane switching (IPS):
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- Twisted nematic (TN):
 - Cheap
 - Poor viewing if not standing directly in front of screen
- Vertical alignment (VA):
 - Good color coverage
 - Higher latency
- Organic light-emitting diode (OLED):
 - No backlight
 - Better contrast
 - flexible

! Question 6 - Where are Wi-Fi antennas typically located on a laptop? Why are they typically placed in this location?

On the sides or top of a laptop, since that is where they best receive wifi signals with no interruption

! Question 7 - What is the purpose of a *backlight* in a laptop display?

Shine enough energy through the color layers so we can see it with our eyes

! Question 8 - An *inverter* serves what purpose in electronics?

Converts DC to AC in order to power a fluorescent backlight

! Question 9 - If a laptop screen has a digitizer, what type of input can that device accept?

touch

! Question 10 - What is the primary wired connection type used to connect a laptop to a peripheral (e.g., keyboard, mouse)?

USB

! Question 12 - Define the term proprietary and provide an example of a proprietary wired connector for mobile devices.

Specific to a company, such as lightning

! Question 13 - Label the connector types pictured below:

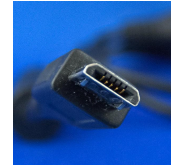
- Connector A: Serial connector
- Connector B: Lightning
- Connector C: Micro - B
- Connector D: Type-A
- Connector E: USB-C



Connector A



Connector B



Connector C



Connector D



Connector E

! Question 14 - What are the advantages of USB Type C connector over USB Type A?

Smaller than some other connectors and can be used in devices that don't have other connector types

! Question 15 - What is NFC (Near Field Communication) commonly used for?

Apple pay

! Question 16 - What is bluetooth?

Wireless communication protocol

! Question 17 - What is the difference between a *touch pen* and an *active stylus*?

Touch pen is cheap and has a conductive tip which is similar to our finger
Active stylus has a conductive tip but also sends information to the device such as pressure and

! Question 18 - What problem does a *docking station* solve?

Solves the problem that laptops do not have the same connectors that PCs have

!/? Question 19 - What does a *port replicator* do? How is it different from a *docking station*?

Docking station uses a proprietary connector, has connectors that you would find on a typical pc and charges your laptop. Also can use expansion cards.

Port replicator connects a single cable to your laptop, so you can't use expansion cards.

Part 2: Mobile Devices - Accessories, Connectivity & Configuration

!/? Question 20 - What are the differences between WiFi, bluetooth, and cellular?

Wifi supports high data transfer

Bluetooth has low data transfer, but allows communication between devices\

Cellular allows internet use and communication over long distances

!/? Question 21 - What use cases were GSM and CDMA originally designed for?

GSM was originally made for voice communications, and CDMA was made so devices could transmit simultaneously

!/? Question 22 - What is the main difference between how GSM and CDMA work? How do they handle multiple calls/users?

GSM each device got a slice of time to transmit

CDMA each phone to transmit individually

!/? Question 23 - What is the purpose of a SIM card?\

Store the individual data of a user to a device.

!/? Question 24 - Briefly explain the evolution of cellular standards from 2G to 5G. Include the change in download rates for each standard.

2G networks 56 kbps-384 kbps has low data throughput, and was developed for voice communications

3G 5mbps-43mbps, could use gps and streaming. Introduces late 1990s

4G 5mbps-150mbps, introduces in 2010s

5G 100mbps-1000mbps, higher bandwidth to use more devices, introduced in 2020

!/? Question 25 - What does *discoverable mode* mean for a bluetooth device?

That another device can connect to it

!/? Question 26 - A device needs to be able to receive signals from at least ___ satellites to determine its longitude latitude, and altitude.

4

!/? Question 27 - If a device cannot reach any GPS satellites, what other technologies can it use to determine its location?

Approximate distance from cell towers or wifi access points

!/? Question 28 - Why should a company use a MDM (Mobile Device Management) solution? What control does it give the company over the device?

disable cameras, restrict apps, enforce password. It is good because they can set policies for company owned devices

!/? Question 29 - List some types of data you can sync between a mobile device and your email provider (e.g., Office 365, Google Workspace, iCloud)?

Mail, photos, calendar, contacts

! Question 30 - Why would we want to synchronize data across different devices?
Provide specific examples.

You could synchronize a company calendar so that all employees have the same meeting times and there is no room for errors or miscommunications between scheduling.